

## Practical week-2

The image displays two screenshots of the MySQL Local application interface, showing SQL queries being executed in the Query 1 tab.

**Top Screenshot:**

```
1 • USE bookflow_db;
2 • CREATE TABLE Loans (
3     loan_id INT PRIMARY KEY,
4     member_id INT,
5     book_id INT,
6     loan_date DATE,
7     FOREIGN KEY (member_id) REFERENCES Members(member_id),
8     FOREIGN KEY (book_id) REFERENCES Books(book_id)
9 );
10 • INSERT INTO Loans
11     (loan_id, member_id, book_id, loan_date)
12     VALUES
13     (1, 1, 1, '2025-01-05'),
14     (2, 2, 2, '2025-01-08'),
15     (3, 3, 3, '2025-01-10'),
16     (4, 1, 2, '2025-02-01'),
17     (5, 2, 1, '2025-02-05'),
18     (6, 3, 2, '2025-02-12'),
19     (7, 1, 3, '2025-03-01'),
20     (8, 2, 3, '2025-03-07'),
21     (9, 3, 1, '2025-03-15'),
22     (10, 1, 1, '2025-04-01');
23 • SELECT * FROM Loans;
```

**Bottom Screenshot:**

```
22
23 • SELECT * FROM Loans;
24 • SELECT
25     m.full_name AS Member_Name,
26     b.title AS Book_Title
27 FROM Loans l
28 INNER JOIN Members
29     ON l.member_id = m.member_id
30 INNER JOIN Books b
31     ON l.book_id = b.book_id;
32 • SELECT
33     published_year,
34     COUNT(book_id) AS Total_Books
35 FROM Books
36 GROUP BY published_year
37 ORDER BY published_year;
38 • CREATE TABLE Donation_History (
39     donation_id INT PRIMARY KEY,
40     book_id INT,
41     donor_name VARCHAR(100),
42     donation_date DATE,
43     FOREIGN KEY (book_id) REFERENCES Books(book_id)
44 );
45 • START TRANSACTION;
46
47 • INSERT INTO Books
48     (book_id, title, isbn, published_year)
49     VALUES
50     (4, 'Animal Farm', '9780451526342', 1945);
51
52 • INSERT INTO Donation_History
53     (donation_id, book_id, donor_name, donation_date)
54     VALUES
55     (1, 4, 'Raj Kumar', CURDATE());
56
57 • COMMIT;
58 • SELECT * FROM Books;
59 • SELECT * FROM Donation_History;
60 • START TRANSACTION;
61
62 • INSERT INTO Books
63     (book_id, title, isbn, published_year)
64     VALUES
65     (5, 'Test Book', '1111111111111', 2020);
66
67 • ROLLBACK;
68 • SELECT * FROM Books;
69 • CREATE INDEX idx_books_isbn
70     ON Books(isbn);
71 • SELECT *
72     FROM Books
73     WHERE isbn = '9780132350884';
74
```

Outputs:



MySQL Local - Warning - not supported

Administration Schemas Query 1 SQL File 5\*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variable
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

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48 (book_id, title, isbn, published_year)
49 VALUES
50 (4, 'Animal Farm', '9780451526342', 1945);
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52 • INSERT INTO Donation_History
53 (donation_id, book_id, donor_name, donation_date)
54 VALUES
55 (1, 4, 'Raj Kumar', CURDATE());
56
57 • COMMIT;
58 • SELECT * FROM Books;
59 • SELECT * FROM Donation_History;
```

100% 32:59

Result Grid Filter Rows: Search Edit: Export/Import:

| donation_id | book_id | donor_name | donation_date |
|-------------|---------|------------|---------------|
| 1           | 4       | Raj Kumar  | 2020-09-19    |

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Local - Warning - not supported

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```

100% 21:48

Result Grid Filter Rows: Search Edit: Export/Import:

| book_id | title         | isbn          | published_year |
|---------|---------------|---------------|----------------|
| 1       | The Alchemist | 9780061122415 | 1988           |
| 2       | Clean Code    | 9780132350884 | 2008           |
| 3       | Atomic Habits | 9780735211292 | 2018           |
| 4       | Animal Farm   | 9780451526342 | 1945           |

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68 • SELECT * FROM Books;
69 • CREATE INDEX idx_books_isbn
70 ON Books(isbn);
71 • SELECT * FROM Books;
```

100% 30:73

Result Grid Filter Rows: Search Edit: Export/Import:

| book_id | title      | isbn          | published_year |
|---------|------------|---------------|----------------|
| 2       | Clean Code | 9780132350884 | 2008           |

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